

A Y Deepesh Raj

Chennai, Tamil Nadu • deepeshrajay@gmail.com • +91 9884539119 • LinkedIn • Github • Portfolio

Professional Summary

Versatile developer with expertise in web development, Python, AI, ML, DL, and data structures. Proficient in Web Development and Python. Skilled in applying AI/ML to solve complex problems. Strong collaborator with a proven ability to deliver high-quality results in fast-paced environments.

Education

Vellore Institute of Technology, Chennai <i>B.Tech in Computer Science and Engineering (CSE) - CGPA:8.29</i>	<i>Sept 2022 – July 2026</i>
Alwin Memorial Public School, Chennai <i>High School Diploma - CGPA:8.54</i>	<i>June 2008 – June 2022</i>

Experience

AI Developer Intern – Forage AI	<i>Feb 2026 – Present</i>
◦ Building an autonomous test setup system to automate end-to-end test cycles and developing an enterprise-ready reinforcement learning framework for prompt optimization using human feedback, PPO, LLM-backed environments, and scalable CI/CD-integrated pipelines.	
Software Engineer Intern – FLSmidth	<i>June 2024 – Sept 2024</i>
◦ Applied 3D reconstruction and image processing to analyze mining equipment degradation, developing OpenCV-based segmentation pipelines that improved defect detection accuracy and inspection efficiency.	

Patents & Publications

• An Intelligent IoT-Based Unauthorized Animal Detection and Classification System	2025
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Certifications & Achievements

• AWS Certified Cloud Practitioner, Solutions Architect & Developer – Udemy	2025
• Semifinalist – Raptree CodeVolt'25 Hackathon (EVXpert AI App)	2025
• GPT4o: Code & Conquer	2024

Projects

RepoMind – GraphRAG-Based AI System for Codebase Intelligence	github.com
◦ Built an AI-powered codebase intelligence tool using Graph-RAG, combining FAISS vector search with call-graph traversal for natural-language Q&A over repositories. Engineered a Tree-sitter ingestion pipeline extracting symbols, call graphs, and dataflow artifacts across multiple languages. Integrated static code health analysis and contributor tracking with a FastAPI + React full-stack interface. Tech Stack: Python, FastAPI, React, LangChain, FAISS, Groq LLM, Tree-sitter, Graph-RAG.	
EduCurate – AI-Driven Personalized Learning System	github.com
◦ Built a modular AI system that generates personalized learning roadmaps, Bloom-level assessments, and automated evaluations. Implemented agentic roadmap generation (topic decomposition, prerequisite DAGs, difficulty scoring, personalization). Developed dual question-generation pipelines using WhisperX+Gemini and T5 transformers with Bloom taxonomy mapping. Integrated a multi-agent Auto-Eval system for semantic scoring, rubric-based grading, and adaptive feedback. Tech Stack: Python, WhisperX, Gemini 2.5 Pro, T5, NLP, DAGs, Multi-Agent Systems.	
Smart Safe Driving App	github.com
◦ Developed a real-time AI-driven driver monitoring system to enhance road safety, using Eye Aspect Ratio (EAR) for drowsiness detection, YOLOv4 for phone usage monitoring, and FER for emotion recognition, integrated with a Flask dashboard and audio alerts. Tech Stack: Python, Flask, OpenCV, dlib, YOLOv4, FER, pyttsx3	
MeetMate – AI Meeting Assistant	github.com
◦ AI tool to transcribe meetings, generate MoMs, and auto-schedule follow-ups into Google Calendar. Combines diarized speech recognition with GPT-based summarization and real-time meeting analysis. Tech Stack: Azure AI Speech Services, OpenAI GPT-4o, Google Cloud APIs, Python, Streamlit, Tailwind CSS	

Technical Skills

Programming and Web: Python, Java, C/C++, SQL, Git, HTML, CSS, Figma, Power BI, Tableau
AI/ML & GenAI: Machine Learning, Transformers, NLP, LangChain, LLM Agents, Topic Modeling, Explainable AI, Exploratory Data Analysis
Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, Numpy, Matplotlib.
Tools & Databases: AWS, Redis, Render, Azure AI, Jupyter Notebook, Git, GitHub, Figma, Vercel, Streamlit, MySQL.
Software Engineering: Data Structures, Algorithms, Software Testing, Agile Methodology, CI/CD Pipeline